

Lecture 1

Embedded System

- It is an arrangement in which all its units assembled work together according to set of rules

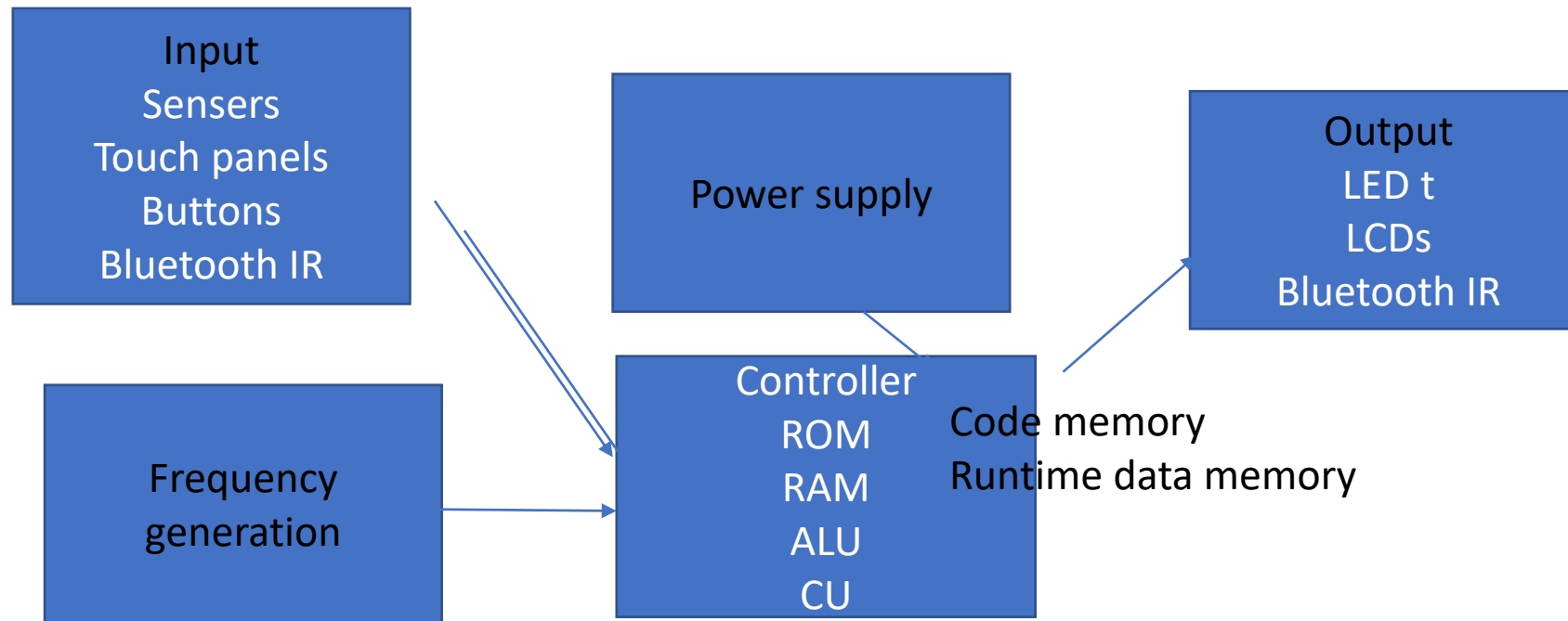
EWatch is a time displaying system

Embedded System

- It is a combination of software and H/W
- It is designed to perform a particular task
- The task has to be completed in a given time.

Eg Washing machine, Micro oven and etc

Building blocks of ES



Applications of Embedded System

- Medical System pace-maker patient monitor system
- Office Equipment Printer and copier fax
- Tool multimeters oscilloscope GPS
- Banking ATMs statement printers
- Transportation Planes, Trains, Boats,
- Automobiles: Engine management trip computer car alarms immobilized airbags ABS
- Building systems: Elevator heater air conditions lighting keycard entries lock
- Agriculture: Feeding system

Characteristics of Embedded System

- Sophisticated functionality
- Realtime operation operation has to be completed by deadline
- Low manufacturing cost
- Low power consumption
- Application dependent processor and not general-purpose processor which we find in computers
- Restricted Memory

General Purpose Computer Vs Embedded System

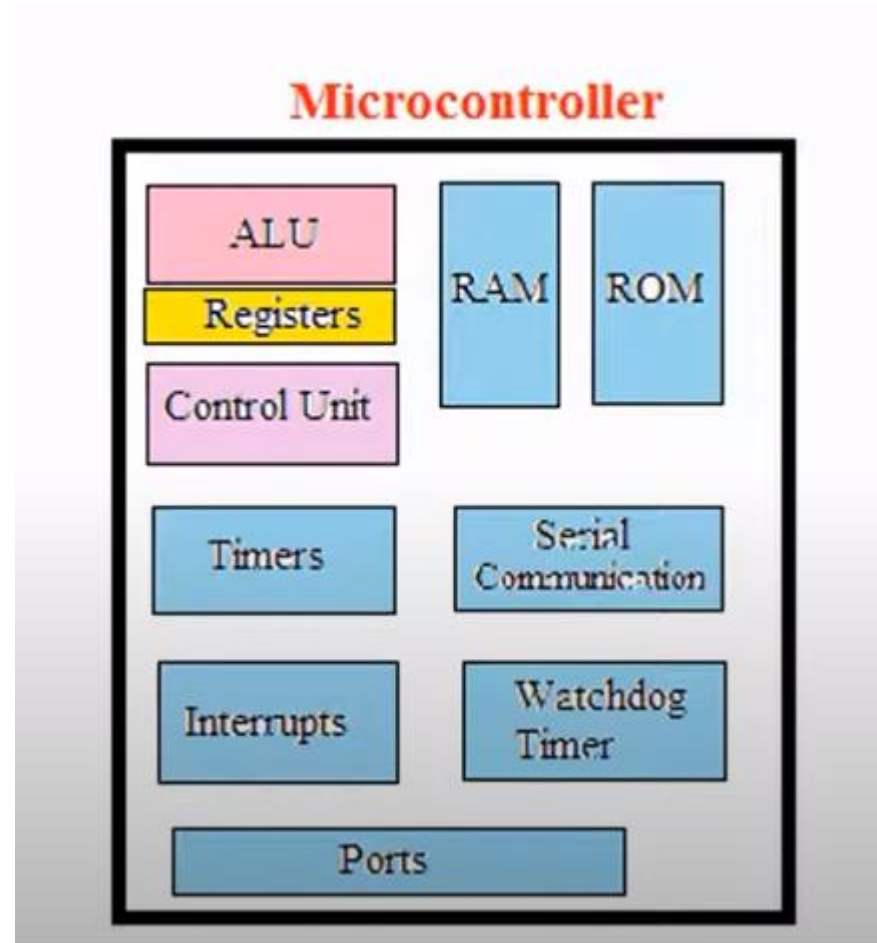
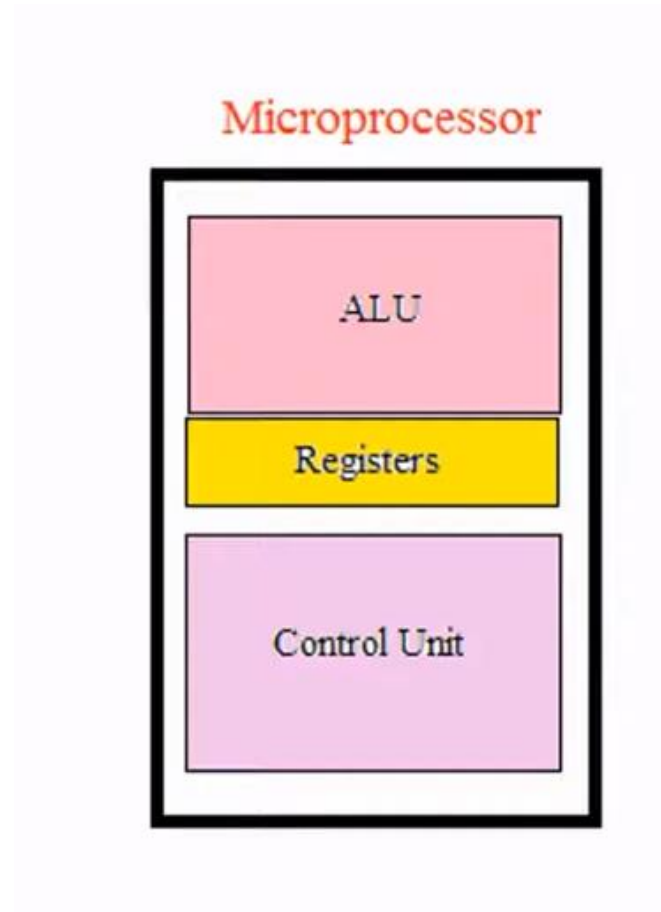
• Purpose	Multipurpose	Single functioned
• Constrained	Low or no resource constrained	Size, Power, cost, Memory, Real time
• Performance	Faster and better	Fixed runtime requirement
• Interface	Keyboard, Display, Mouse, Touch screen	Integrated into the real world Button sensors

Microprocessor and Microcontroller

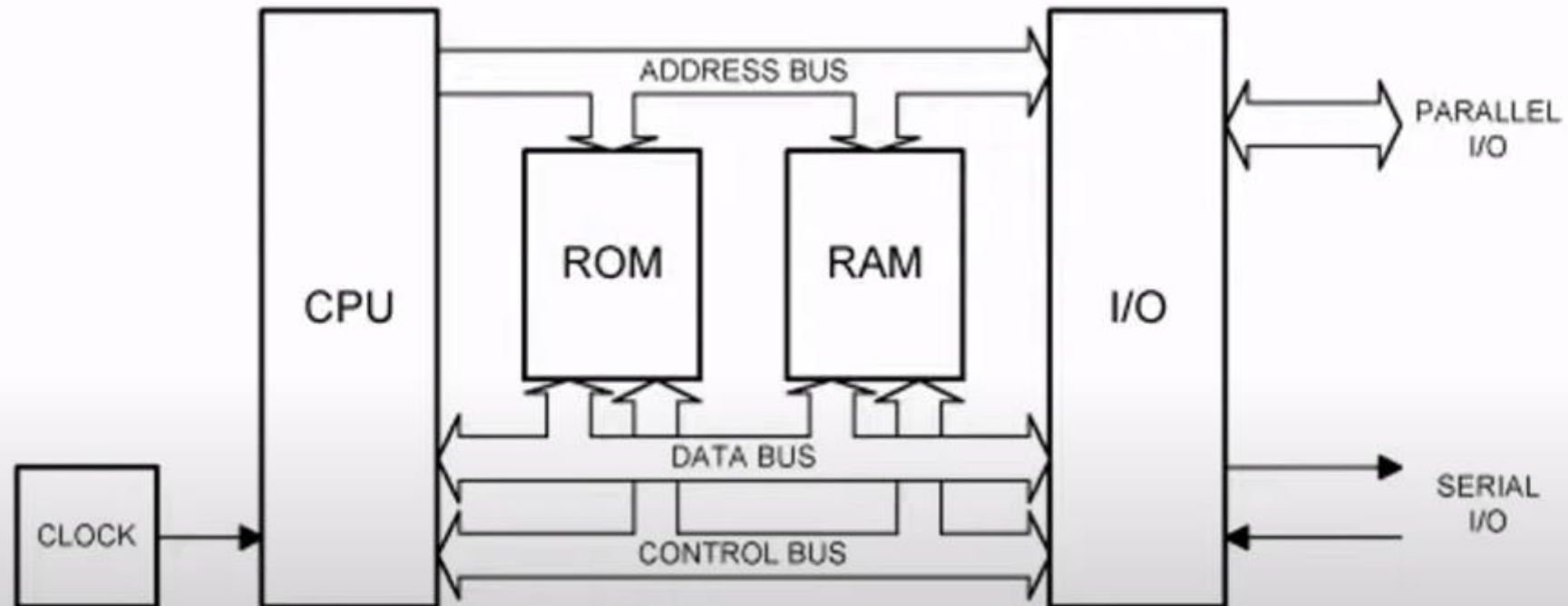
- Digital Device
- Arithmetic and Logical Operation
- Instructions given by users

Difference between Microprocessor and Microcontroller

Block Diagram



Microprocessor Based System



Difference between Microprocessor and Microcontroller

- Microprocessor based systems are more flexible
- Microcontroller based system are less flexible
- Microprocessors are used for general purpose applications

Ex Desktop PC Laptop notepads

- Microcontrollers are designed to perform specific tasks. Where some controlling actions are required

Ex keyboards printers, mouse, washing machine, digicam and pen drive

Von Neumann Vs Harvard

- Types of architecture

-

Von Neumann: data and code lie in same memory block

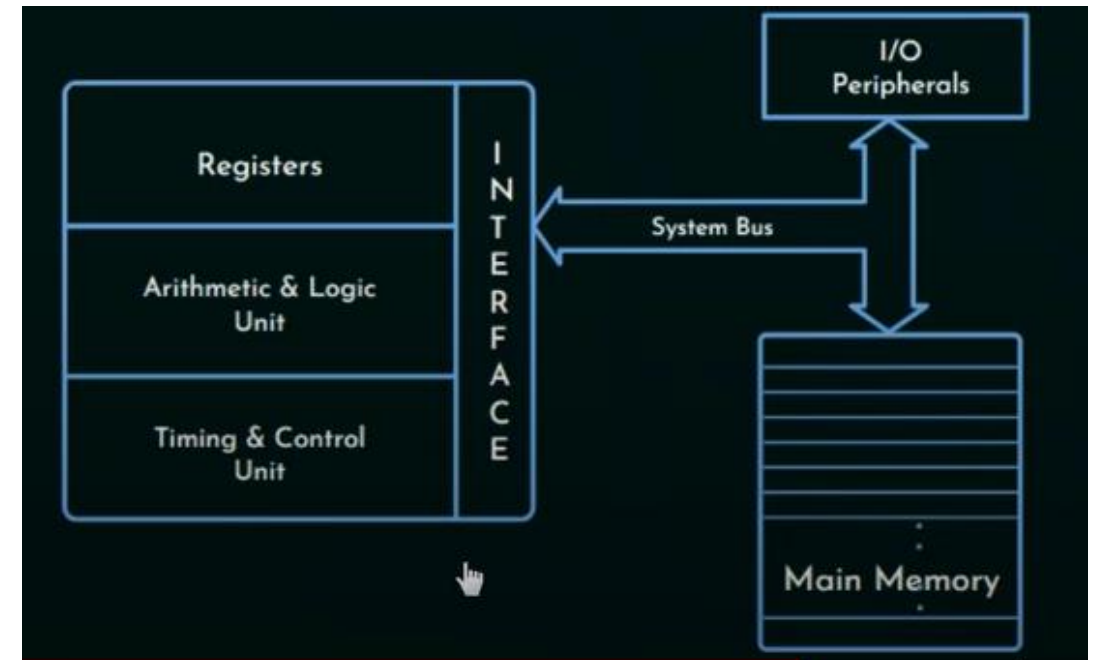
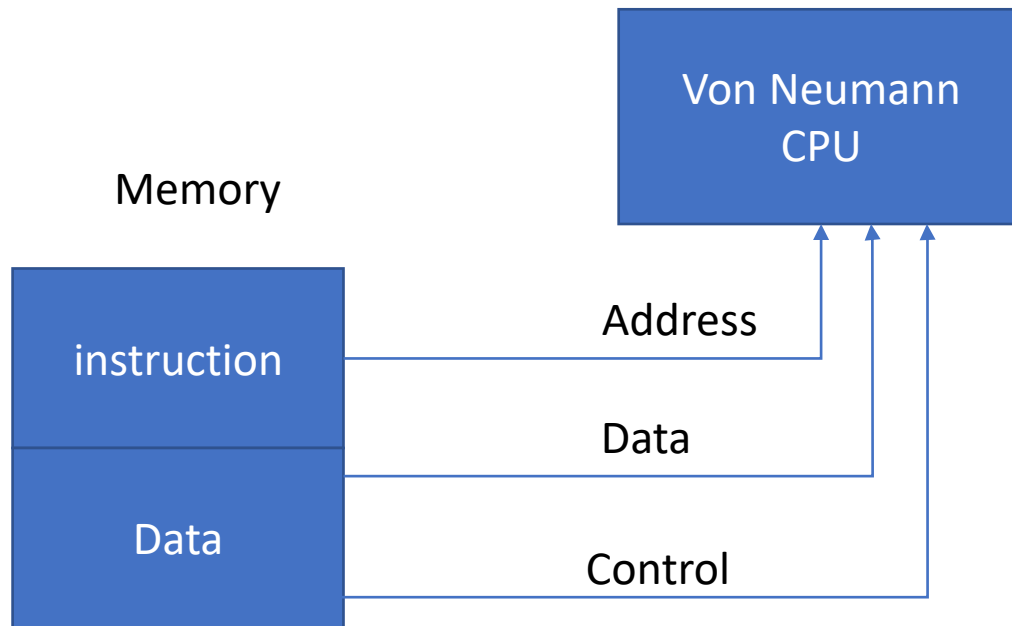
- 1 data bus used for both instruction and data
- 1. CPU can be performed only one operation at a time
- 2 set of clocks cycles required 1 for data fetch and 1 for instruction fetch
- Pipelining is not possible
- Simple in design

- Harvard: data and code lie in different memory block

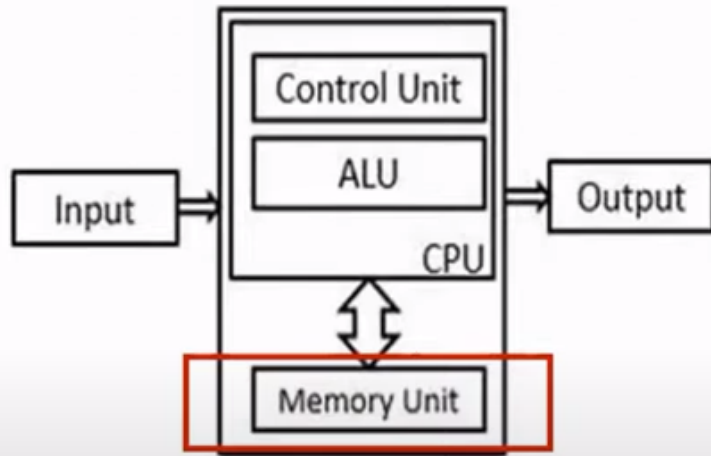
- Separate memory for code and data
- Single clocks cycle for data fetch and instruction fetch
- Pipelining is possible
- Complex in design

Von Neumann Vs Harvard

- Von Neumann

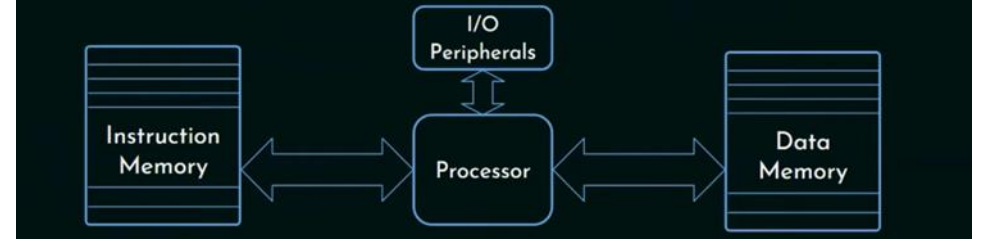


•Von Neumann Architecture

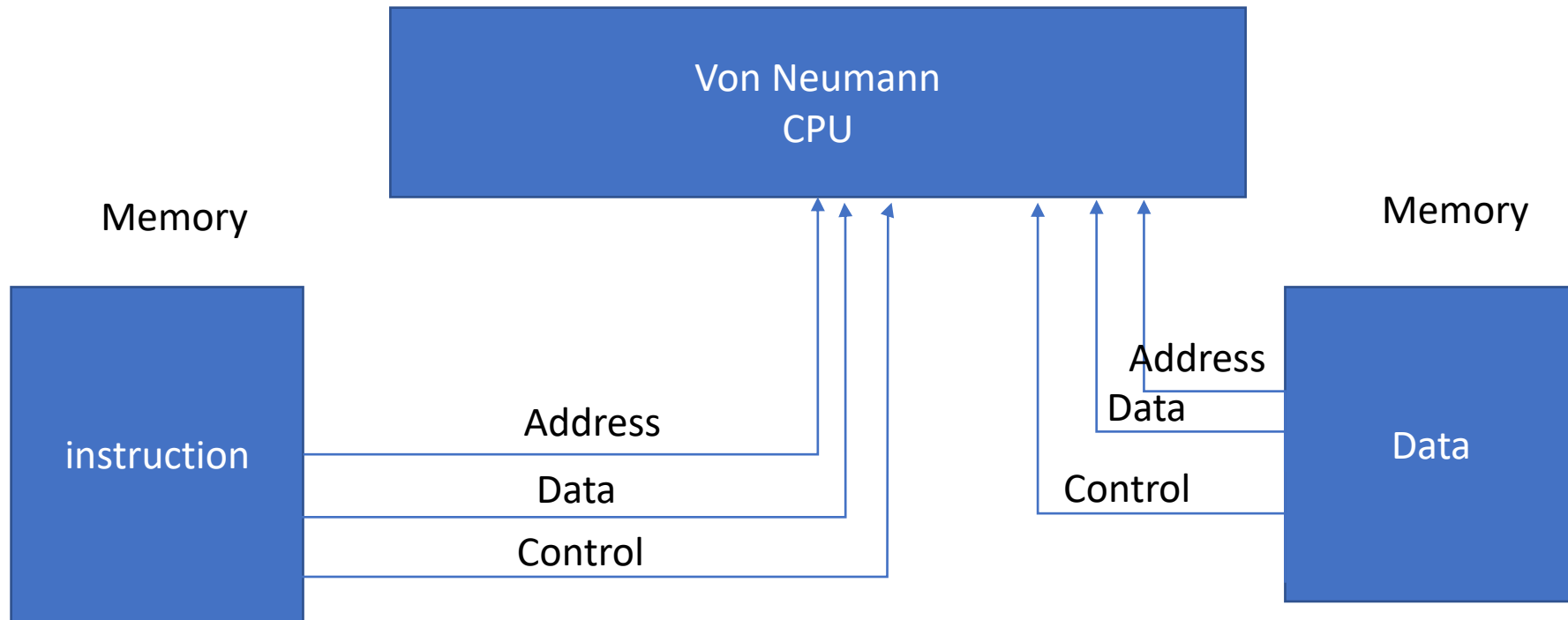


- Named after the computer scientist John Von Neumann.
- The computer has single memory for storing data as well as program.
- Von Neumann architecture has only one bus for instruction and data.
- Design of the von Neumann architecture is simple.
- It is cheaper.

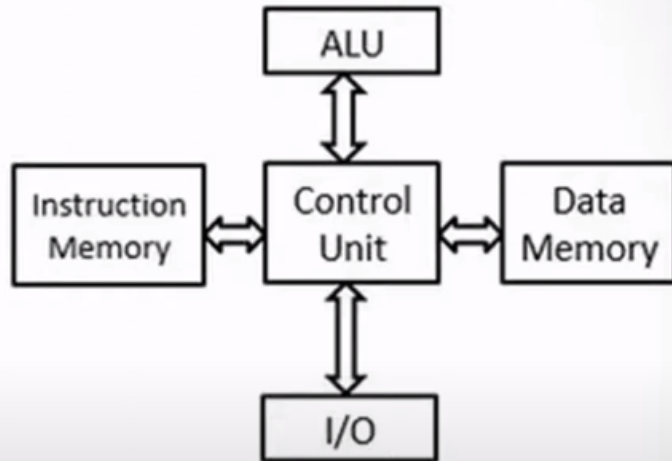
Von Neumann Vs Harvard



- Harvard Architecture



•Harvard Architecture

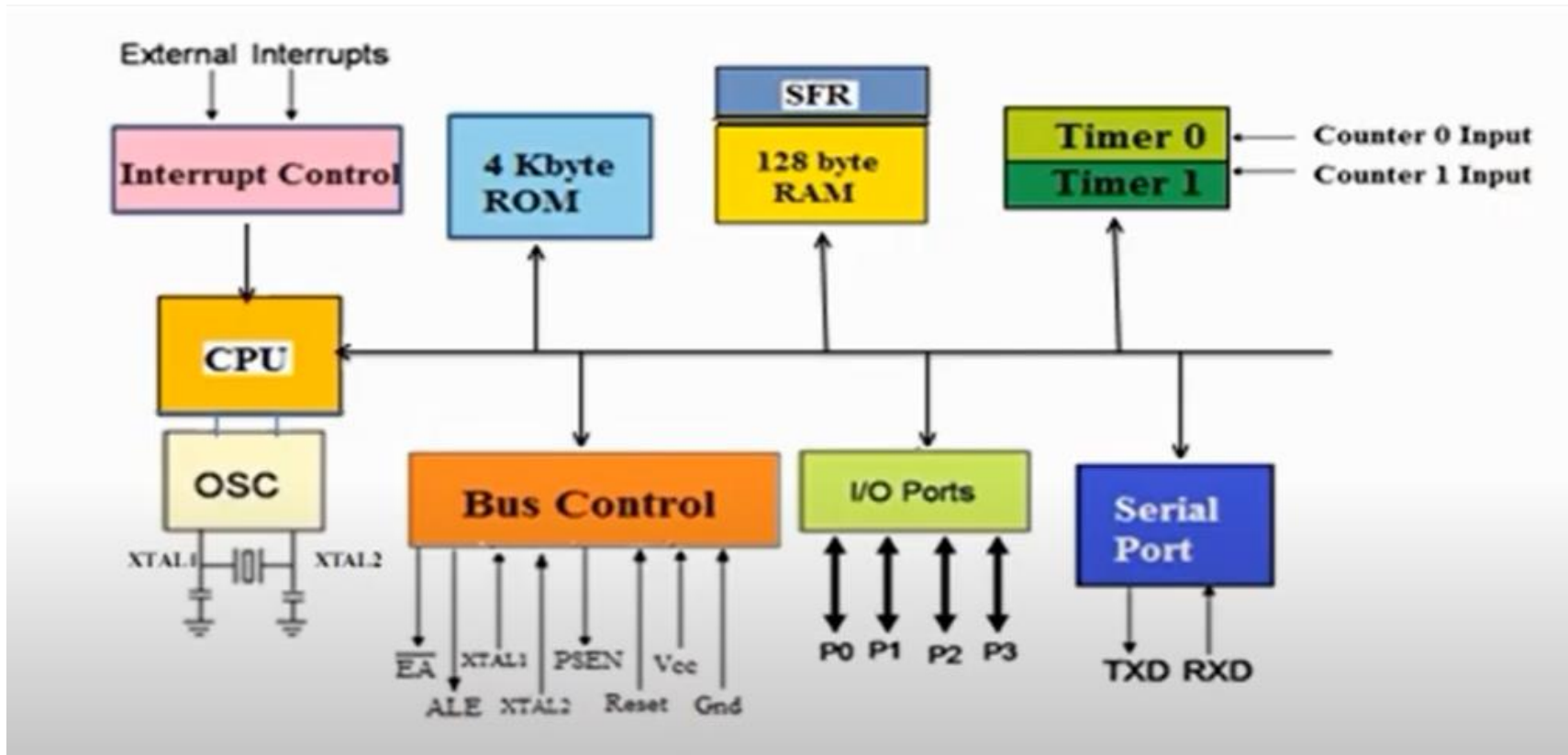


- The computer has two separate memories for storing data and program.
- Harvard architecture is required separate bus for instruction and data.
- Most of the modern computing architectures are based on Harvard architecture.
- Design of Harvard architecture is complicated.
- Comparatively high cost.

RISC and CISC

RISC	CISC
Stands for “Reduced Instruction Set Computing”.	Stands for “Complex Instruction Set Computing”.
It has less number of instructions and a comparatively smaller instruction set.	It has more instructions and a more complex instruction set.
Less addressing modes.	More addressing modes.
Fixed instruction size and hence each instruction takes one machine cycle to execute.	Each instruction can take 2-10 machine cycles.
Simple hardware, lower manufacturing costs.	Complex hardware and higher costs.
Low power consumption.	High power consumption.
PIC, ARM.	x86, 8051

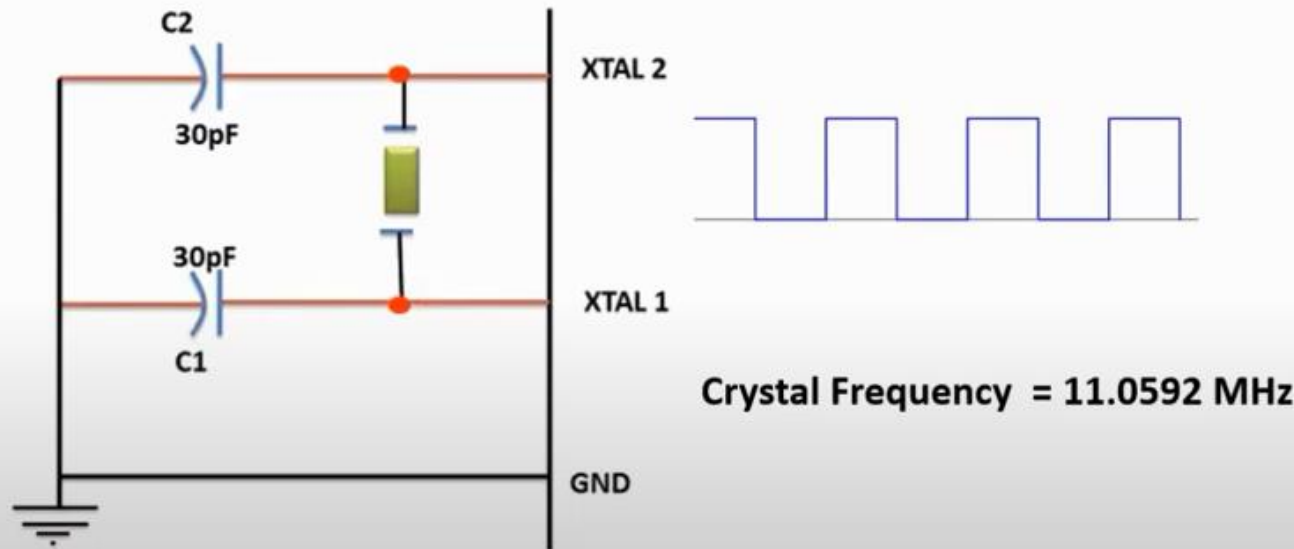
Architecture of 8051 Microcontroller



8 bit microcontroller
Internal rom 4 kbyte
000H to FFFH
Internal RAM 128 byte
00 Hto 7FH
Two timer for timing delays
4 General Input port P0to
P3Internal Serial Port TXD and
RXD

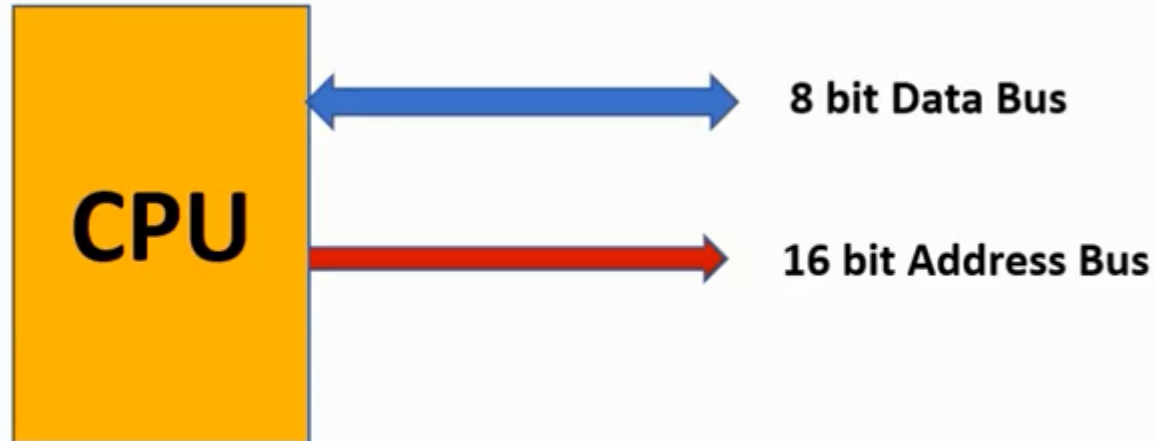
Feature of 8051

- Operating frequency range is from 1 MHz to 16MHz.



Feature of 8051

- Data bus size is 8 bit



- Address bus size is 16 bit

Maximum memory size is 2^{16} or 64 Kbytes

Feature of 8051

- Internal ROM memory size is 4 Kbyte



- Internal RAM memory size is 128 bytes

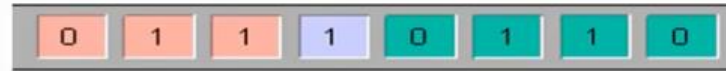


- It has special function registers

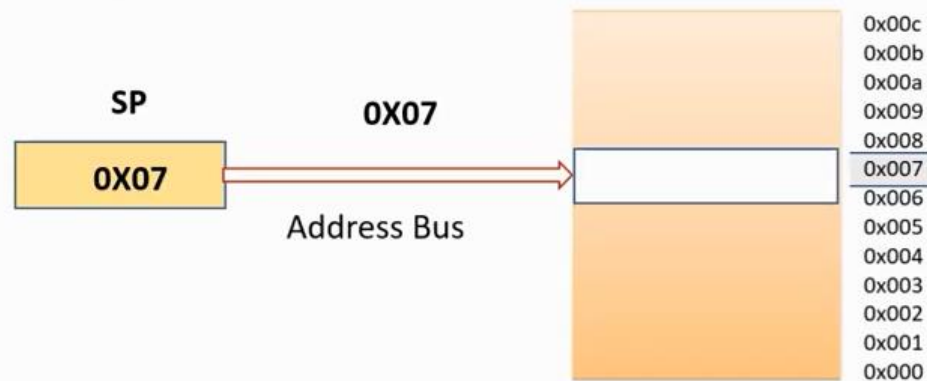


Feature of 8051

- 8 bit ALU having registers **A (8 bit)** which is called as an **accumulator** and register **B (8 bit)** called as math register.

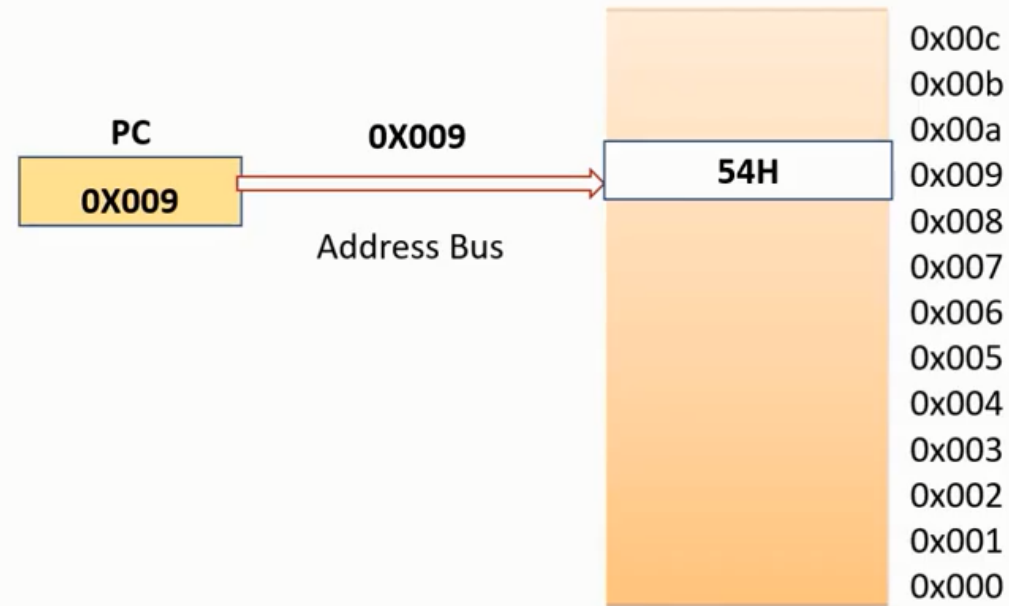


- 8 bit Program status word register (PSW)
- 8 bit stack pointer register (SP)



Stack memory is in Internal RAM and the top of stack is address of 07H

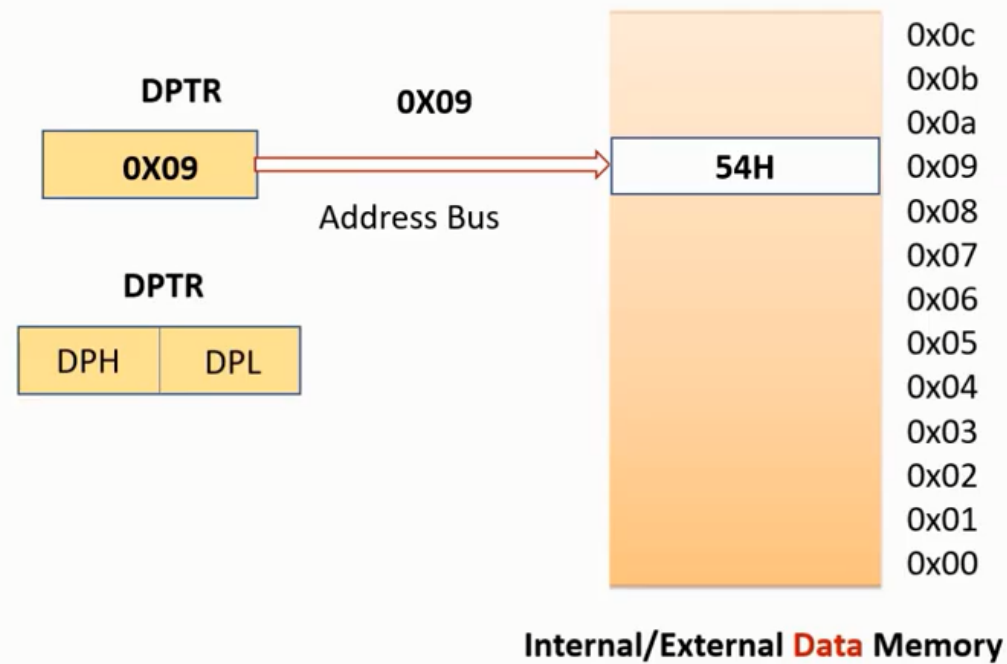
- 16 bit program counter register (PC)



Internal/External **Program** Memory

Feature of 8051

- 16 bit Data pointer register (DPTR)



Feature of 8051

- **Two, 16 bit timers/counters**
- **5 interrupts +1 Reset**
- **It has Universal asynchronous receiver transmitter (UART) block for serial communication.**
- **4 input/output ports. Each port is of 8 bit.**

Various Versions of 8051

Intel fabricated the **original 8051** which is known as MCS-51. The other two members of the 8051 family are 8052 and 8031:

Features	8051	8052	8031
RAM(bytes)	128	256	128
ROM	4K	8K	0K
Timers	2	3	2
Serial port	1	1	1
I/O pins	32	32	32
Interrupt sources	6	8	6

AT89C51 from Atmel Corporation

Part Number	ROM	RAM	I/O pins	Timer	Interrupt	Vcc	Packaging
<u>AT89C51</u>	4K	128	32	2	6	5V	40
<u>AT89C52</u>	8K	256	32	3	8	5V	40
AT89C1051	1K	64	15	1	3	3V	20
AT89C2051	2K	128	32	3	8	3V	20
AT89LV51	4K	128	32	2	6	3V	40
AT89LV52	8K	128	32	3	8	3V	40

DS5000 from Dallas Semiconductor

Part Number	RAM	ROM	Timers	I/O pins	Interrupts	Vcc	Packaging
DS5000-8	128	8K	2	32	6	5V	40
DS5000-32	128	32K	2	32	6	5V	40
DS5000T-8	128	8K	2	32	6	5V	40
DS5000T-8	128	32K	2	32	6	5V	40

Format of Program Status Word Register

D7	D6	D5	D4	D3	D2	D1	D0
CY	AC	F0	RS1	RS0	OV	-	P

CY Carry Flag

AC Auxiliary Carry Flag

F0 Flag 0 available to user for general purpose.

RS1 Register Bank selector bit 1

RS0 Register Bank selector bit 0

OV Overflow Flag

- Reserved

P Parity FLAG. Set/ cleared by hardware during instruction cycle to indicate even/odd number of 1 bit in accumulator.

$$\begin{array}{r} 11110101 \\ + 10001010 \\ \hline 10111111 \\ \text{CY} \end{array}$$

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$$\begin{array}{r} \text{CY} \\ 1 \\ 1101 \quad 1101 \\ + \quad 0000 \quad 1010 \\ \hline 0111 \quad 0111 \end{array}$$

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RS1	RS0	Register Bank
0	0	0
0	1	1
1	0	2
1	1	3

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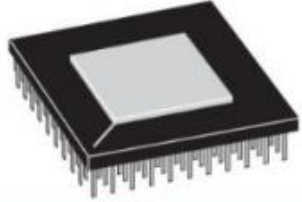
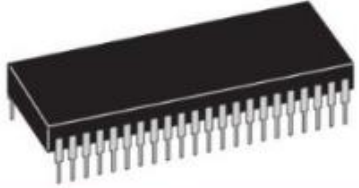
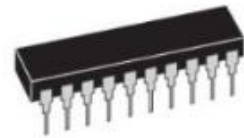
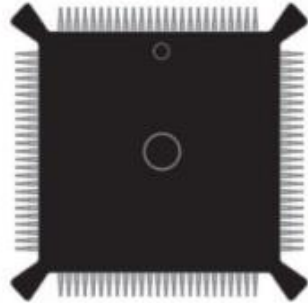
P Parity FLAG. Set/ cleared by hardware during instruction cycle to indicate even/odd number of 1 bit in accumulator.

1 0 0 1 0 1 0 0 P=1

1 1 0 1 0 1 0 0 P=0

Packaging Types

8051 Microcontroller Packaging Types

Packages		
CPGA Ceramic through-hole package 	SDIP Plastic through-hole package 	HDIP Plastic through-hole package with higher heat dissipation rate 
PLCC Plastic leaded chip carrier 	QFP Quad flat package 	HSOP Plastic dual-construction surface mount package with higher heat dissipation rate 